Jiawei Xu

Ph.D. Student in Information Studies

iSchool, School of Computing, The University of Texas at Austin

Website Google Scholar GitHub LinkedIn OpenReview

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I study the foundations of AI agents, with applications in health.

Education

Ph.D. Student in Information Studies Aug. 2023–Present

The University of Texas at Austin
iSchool, School of Computing

Committee: Professor Ying Ding (Chair), Professor Yan Zhang, and Professor Kenneth Fleischmann.

Master of Science Sep. 2021–Jul. 2023

Information Science
Peking University

Advisors: Professor Yi Bu and Professor Win-bin Huang.
Excellent Graduate of Beijing

Bachelor of Management Sep. 2017–Jul. 2021

Information Management and Information Systems
Peking University

Publications

Research themes: LLM / agents | Health | Science of Science. Multiple tags indicate overlapping themes.

Conference papers

C1 **SEER: Long-Context Reasoning via Selective Visual-Text Compression**

Xu, J., Zhai, Z., Fang, J., Xu, R., Lu, M., Zhang, Y., Wang, G., Chen, T., & Ding, Y.

2026 / *Conference on Language Modeling (COLM 2026)*. Paper Code

LLM / agents

C2 **MedHallu: A Comprehensive Benchmark for Detecting Medical Hallucinations in Large Language Models**

Pandit, S., **Xu, J.**, Hong, J., Wang, Z., Chen, T., Xu, K., & Ding, Y.

2025 / *EMNLP 2025 Main Conference*. Paper Dataset Code

LLM / agents | Health

C3 **Mapping from Meaning: Addressing the Miscalibration of Prompt-Sensitive Language Models**

Cox, K., **Xu, J.**, Han, Y., Xu, R., Li, T., Hsu, C.-Y., Chen, T., Gerych, W., & Ding, Y.

2025 / *Proceedings of AAAI 2025*. Code Paper

LLM / agents

Publications (continued)

Conference papers (continued)

C4 **From Newborn to Impact: Bias-Aware Citation Prediction**

Lu, M., Wu, M., **Xu, J.**, Li, W., Liu, F., Ding, Y., Sun, Y., Lu, J., & Zhang, Y

2026 / *The Web Conference 2026*. Preprint

LLM / agents | Science of Science

C5 **Interactive Graph Visualization and Teaming Recommendation in an Interdisciplinary Project's Talent Knowledge Graph**

Xu, J., Chen, J., Ye, Y., Sembay, Z., Thaker, S., Payne-Foster, P., Chen, J., & Ding, Y.

2025 / *ASIS&T 2025*. Demo

LLM / agents | Health | Science of Science

C6 **Path-RAG: Knowledge-Guided Key Region Retrieval for Open-ended Pathology Visual Question Answering**

Naeem, A., Li, T., Liao, H.-R., **Xu, J.**, Mathew, A. M., Zhu, Z., Tan, Z., Jaiswal, A. K., Salibian, R. A., Hu, Z., Chen, T., & Ding, Y.

2024 / *ML4H 2024*. Code Paper

LLM / agents | Health

C7 **Decoding Patterns of Data Generation Teams for Clinical and Scientific Success: Insights from the Bridge2AI Talent Knowledge Graph**

Xu, J., Xie, Q., Liu, M., Sembay, Z., Thaker, S., Payne-Foster, P., Chen, J. Y., & Ding, Y.

2024 / *Proceedings of JCDL 2024*.

LLM / agents | Health | Science of Science

C8 **Thought Graph: Generating Thought Process for Biological Reasoning**

Hsu, C.-Y., Cox, K., **Xu, J.**, Tan, Z., Zhai, T., Hu, M., Pratt, D., Chen, T., Hu, Z., & Ding, Y.

2024 / *Companion Proceedings of the ACM Web Conference 2024 (WWW'24 Companion)*. Paper

LLM / agents | Health

Journal articles

J1 **Knowledge Integration and Diffusion Structures of Interdisciplinary Research: A Large-Scale Analysis Based on Propensity Score Matching**

Xu, J., Zheng, Z., Min, C., Huang, W., & Bu, Y.

2025 / *Journal of the Association for Information Science and Technology*. Code Paper

Science of Science

J2 **TAMA: A Human-AI Collaborative Thematic Analysis Framework Using Multi-Agent LLMs for Clinical Interviews**

Xu, H., Yi, S., Lim, T., **Xu, J.**, Well, A., Mery, C., Zhang, A., Zhang, Y., Ji, H., Pingali, K., Leng, Y., & Ding, Y.

2026 / *ACM Transactions on Computing for Healthcare*. Paper

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Publications (continued)

Journal articles (continued)

J3 **PubMed Knowledge Graph 2.0: Connecting Papers, Patents, and Clinical Trials in Biomedical Science**

Xu, J., Yu, C., **Xu, J.**, Torvik, V. I., Kang, J., Sung, M., Song, M., Bu, Y., & Ding, Y.

2025 / *Scientific Data*.

Health | Science of Science

Workshop proposal

2025 **The Second Workshop on GenAI for Health: Potential, Trust, and Policy Compliance**

Xu, J., Xiang, T., Rajpurkar, P., Hong, J., Chen, C., Adeli, E., Li, X., Pavlakos, G., Delp, S., Fei-Fei, L., & Ding, Y

2025 / *NeurIPS 2025 Workshop Proposal*.

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Preprints & work under review

P1 **Rethinking the Value of Multi-Agent Workflow: A Strong Single Agent Baseline**

Xu, J., Koesdwiady, A., Bei, S., Han, Y., Huang, B., Wang, D., Chen, Y., Wang, Z., Wang, P., Li, P., & Ding, Y

2026 / *arXiv preprint*. Preprint

LLM / agents

P2 **MATRIX: Multi-Agent Teaming Recommendation through EXpertise Gap Identification**

Xu, J., Ye, Y., Han, Y., Wang, H., Shomer, H., Li, W., Lu, M., Wu, M., Liu, F., Zhang, Y., Sun, Y., Tang, J., & Ding, Y

2026 / *Under review*

LLM / agents | Health | Science of Science

Workshop papers

W1 **Beyond Feature Importance: Auditable Feature Interactions in Predicting Post-Stroke Rigidity with Graph Explainable AI**

Xu, J., Lee, Y., Youssef, A. E., Huang, T., Guo, T., Ying, R., & Ding, Y.

2026 / *DAIH: LLM/VLM Deployment Opportunities and Risks in Healthcare, COLM 2026 Workshop*. Poster. Paper Workshop

Health

W2 **LLM-TA: An LLM-Enhanced Thematic Analysis Pipeline for Transcripts from Parents of Children with Congenital Heart Disease**

Xu, J., Raza, Z., Lim, T., Boddy, L., Mery, C., Well, A., & Ding, Y.

2025 / *GenAI4Health Workshop at AAAI*. Code Paper

LLM / agents | Health

W3 **Position: Open and Closed Large Language Models in Healthcare**

Xu, J., Ding, Y., & Bu, Y.

2024 / *GenAI4Health Workshop at NeurIPS*. Google Scholar

LLM / agents | Health

Employment

May–Aug. 2026

Applied Scientist Intern

Amazon / Seattle, WA

Self-improving automated research loop for machine learning tasks.

Mentors: Dihan Dai and Keyang He; Manager: Jinhe Shi. Technical advisors: Jiajian Lu and Olcay Boz.

May–Nov. 2025

Applied Scientist Intern

Amazon Search / Palo Alto, CA (SFO24)

Rethinking the Value of Multi-Agent Workflow: A Strong Single Agent Baseline. Featured as the cover paper in DAIR.AI Top AI Papers of the Week.

Mentors: Arief Koesdwiady, Sisong Bei, and Yan Han; Manager: Yutong Chen.

Aug. 2023–
Aug. 2026

Graduate Research Assistant

The University of Texas at Austin

iSchool, School of Computing

SELECTED PROJECTS

- **Patient Simulator** Demo

  Developed a voice-based AI patient simulator to help medical students practice clinical conversations and patient communication. Related exploration: SEER, long-context reasoning through selective visual-text compression.

- **Cell Maps for AI (CM4AI)** Demo

  Developed interactive knowledge graph visualization and teaming recommendation tools to help researchers explore expertise and identify potential collaborators.

- **LLM uncertainty & hallucination detection for health**

  Core development contributions to Mapping from Meaning, studying prompt sensitivity and uncertainty calibration, and MedHallu, a benchmark for detecting medical hallucinations.

- **LLM-assisted thematic analysis** Paper Code

  Developed methods for identifying themes in interviews with parents of children with congenital heart disease, with evaluation against human analysis and expert review.

Academic service

Workshop leadership

2026 **Workshop Organizer (Student Lead) & Program Co-Chair**
GenAI4Health Workshop at NeurIPS 2026
Sydney, Australia

2025 **Workshop Organizer (Lead) & Program Co-Chair**
GenAI4Health Workshop at NeurIPS 2025
San Diego, CA, USA

2024 **Onsite Student Organizer**
GenAI4Health Workshop at NeurIPS 2024
Vancouver, Canada

Reviewing

Conference PC member & reviewer

- ICML
- NeurIPS
- ACL Rolling Review
- ML4H
- AAAI

Journal reviewer

- ACM Computing Surveys
- Data and Information Management
- Journal of the Association for Information Science and Technology
- Nature Humanities & Social Sciences Communications
- Online Information Review
- Data Intelligence
- Information & Culture
- Journal of Clinical and Translational Science
- Scientometrics

Invited talks

Oct. 2026	**AI Agents over Biomedical Knowledge Graphs: Finding Collaborators and Querying Data from Research Papers** NSF Convergence Accelerator meeting Project demo
Apr. 2025; Oct. 2024	**CM4AI Talent Knowledge Graph** Virtual talk
Aug. 2023	**Beyond Correlation: What Factors Influence Scientific Performance** Virtual talk

Teaching

2024–2025	**LLM Prompt Engineering and Multi-Agent Systems** Guest lecturer, The University of Texas at Austin
2021–2023	**Data Visualization** Teaching assistant, Peking University
2022	**Complex Networks** Teaching assistant, Peking University

Technical expertise

Languages	English (professional); Mandarin (native).
Everyday workflow	Codex + Claude Code, with me in the loop, taking ideas through research, experiments, and live applications.
Agent harnesses	Harness improvement, skills, prompting, and memory.
LLM post-training	Supervised fine-tuning (SFT) and reinforcement learning (RL) for LLM agents.